pii-steady-aim — the worst-group threshold rule

Run `pii-steady-aim-2026-08-25` · commit `78a8bb1` · 1,000 trials · 16 measured arms (2 models × 8 corpora) · MLflow experiment `pii-steady-aim`

A single-mechanism follow-up to `pii-quiet-alarm`. Everything is carried over unchanged except how per-label and document thresholds are chosen.

Results

Headline — equal-corpus, sealed evaluation

Arm	priority macro F0.5	macro F0.5	doc precision	doc specificity	doc recall	one-core p95	verdict
steady-cascade	0.7474	0.6932	**0.8870**	**0.8798**	0.7582	4.11 ms	**blocked**
champion_1k *(prior)*	0.2057	0.4163	0.6162	0.0005	0.9994	2.03 ms	blocked
quiet-cascade (previous run)	*0.8233*	*0.7766*	*0.8806*	*0.8693*	*0.7729*	*3.93 ms*	*blocked*

Did the fix work?

Yes, for what it targeted — and it cost what the probe said it would.

	pii-quiet-alarm	pii-steady-aim	Δ
priority tag recall pairs passing	36 / 55	**42 / 55**	**+6**
worst doc precision (ci_lower)	0.7893	**0.8024**	+0.0131
worst doc specificity (ci_lower)	0.8299	**0.8486**	+0.0187
worst doc recall (ci_lower)	0.6604	0.6534	–0.0070
priority macro F0.5	0.8233	0.7474	–**0.0759**
priority macro precision	—	0.7520	—

The eleven "marginal" failures the fix was aimed at — pairs sitting between 0.70 and 0.77 against a 0.75 floor — are gone. Every remaining failure moved up substantially:

Pair	pii-quiet-alarm	pii-steady-aim
medical_record_number_mrn @ betterdataai	0.0132	**0.3642**
password @ betterdataai	0.1000	**0.3200**
address @ govdocs2	0.3636	**0.5455**
address @ betterdataai	0.3110	**0.6299**

Pair	pii-quiet-alarm	pii-steady-aim
full_name @ govdocs2	0.6208	0.6053
driver_s_license_number @ openpii	0.6206	**0.6902**

The gate ladder — why nothing was promoted

Per corpus, on the bootstrap lower bound, 30-instance minimum.

Arm	doc precision ≥ 0.90	doc specificity ≥ 0.85	doc recall ≥ 0.85	priority tag recall ≥ 0.75	p95 ≤ 5 ms
steady-cascade	**FAIL** 1/3 (0.8024)	**FAIL** 2/3 (0.8486)	**FAIL** 1/3 (0.6534)	**FAIL** 42/55 (0.2000)	PASS
champion_1k	**FAIL** 0/3 (0.4632)	**FAIL** 0/3 (0.0000)	PASS 3/3	PASS 55/55	PASS

No feasible arm. The artifact was packaged on explicit request as the run's best deliverable and **not promoted to** `@champion` — its model card leads with this table.

Document level, per corpus

Point estimate, bootstrap lower bound in brackets. Previous run in italics.

Corpus	precision	specificity	recall
pii2 (synthetic)	0.9792 (0.9772)	0.8973 (0.8889)	0.9306 (0.9276)
datax (real)	0.8206 (0.8024)	0.8649 (0.8486)	0.6743 (0.6534)
govdocs2 (real)	0.8612 (0.8484)	0.8773 (0.8659)	0.6696 (0.6540)
pii2, previous	*0.9783*	*0.8926*	*0.9324*
datax, previous	*0.8089*	*0.8462*	*0.7104*
govdocs2, previous	*0.8545*	*0.8692*	*0.6759*

Precision and specificity improved on **all three** corpora. Document recall on the two real-world corpora slipped slightly, and it is now the binding failure.

Priority-tag quality, per corpus

Corpus	priority F0.5	precision	recall	n
pii2	0.9296	0.9268	0.9597	30,000
pii_holdout	0.8473	0.8469	0.9141	20,000
openpii	0.8325	0.8388	0.8596	38,937
ai4privacy	0.7707	0.7941	0.7726	10,626

Corpus	priority F0.5	precision	recall	n
betterdataai	0.3570	0.3533	0.5320	10,360

pii2 flatters; betterdataai (silver labels) is the weak corpus. Quote **0.7474**.

TL;DR

- **The fix did exactly what it was designed to do.** Failing priority pairs went 19 → 13, and the marginal cluster it targeted is gone entirely. Worst document precision and specificity both improved.
- **It cost 0.076 priority macro F0.5**, as the pre-launch probe predicted. The 0.75 recall floor is a hard gate, so buying it has to be paid for in precision; the search found the cheapest robustness that passes rather than the safest.
- **Still no feasible arm**, so nothing was promoted. What remains is no longer a calibration-margin problem — `address` (0.55–0.65) and `full_name` (0.61–0.64) are genuinely below the floor across corpora.
- **Packaged anyway, on request**, as `pii-steady-aim-cascade-v1`. It reproduces its recorded metric through its own `tagger.py` to **delta 0.00000**, and the card's first section is the gate table it fails.
- Against the model actually in production: priority macro F0.5 **0.2057** → **0.7474**, document specificity **0.0005** → **0.8798**, at 4.11 ms p95 on one core.

Model scoreboard

Model	Dataset	F1@5	F1@3	F1@1	Docs/sec	ms/doc
`steady-cascade`	10360_betterdataai_ner_silver_eval_10.36k					
`steady-cascade`	10626_ai4privacy_pii_masking_eval_10.63k					
`steady-cascade`	20000_pii_holdout_20.00k					
`steady-cascade`	30000_pii2_eval_25.15k					
`steady-cascade`	38937_openpii_pii_eval_38.94k					
`steady-cascade`	4000_datax-dualjudge-evalset-1.32k					
`steady-cascade`	5617_nemotron_eval_5.36k					
`steady-cascade`	6589_govdocs2-dualjudge-eval20-3.53k					
`champion_1k`	10360_betterdataai_ner_silver_eval_10.36k					
`champion_1k`	10626_ai4privacy_pii_masking_eval_10.63k					
`champion_1k`	20000_pii_holdout_20.00k					

Model	Dataset	F1@5	F1@3	F1@1	Docs/sec	ms/doc
`champion_1k`	30000_pii2_eval_25.15k					
`champion_1k`	38937_openpii_pii_eval_38.94k					
`champion_1k`	4000_datax-dualjudge-evalset-1.32k					
`champion_1k`	5617_nemotron_eval_5.36k					
`champion_1k`	6589_govdocs2-dualjudge-eval20-3.53k					

Measured 2026-08-25.